TiDB

SOLUTION BRIEF

Agentic Coding - Shared Memory Layer

EXECUTIVE SUMMARY

One shared memory for a team of agents.

Bottom line. mem9 gives a team of AI coding agents one shared, queryable memory on TiDB Cloud. Agents read before they build and write back when done - so none starts cold or duplicates work. It is isolated per team, an agent can work across repos in a single session, and it answers the one question a vector store cannot: the transitive blast radius of a change. Every screenshot in this brief is a live read or write against TiDB Cloud.

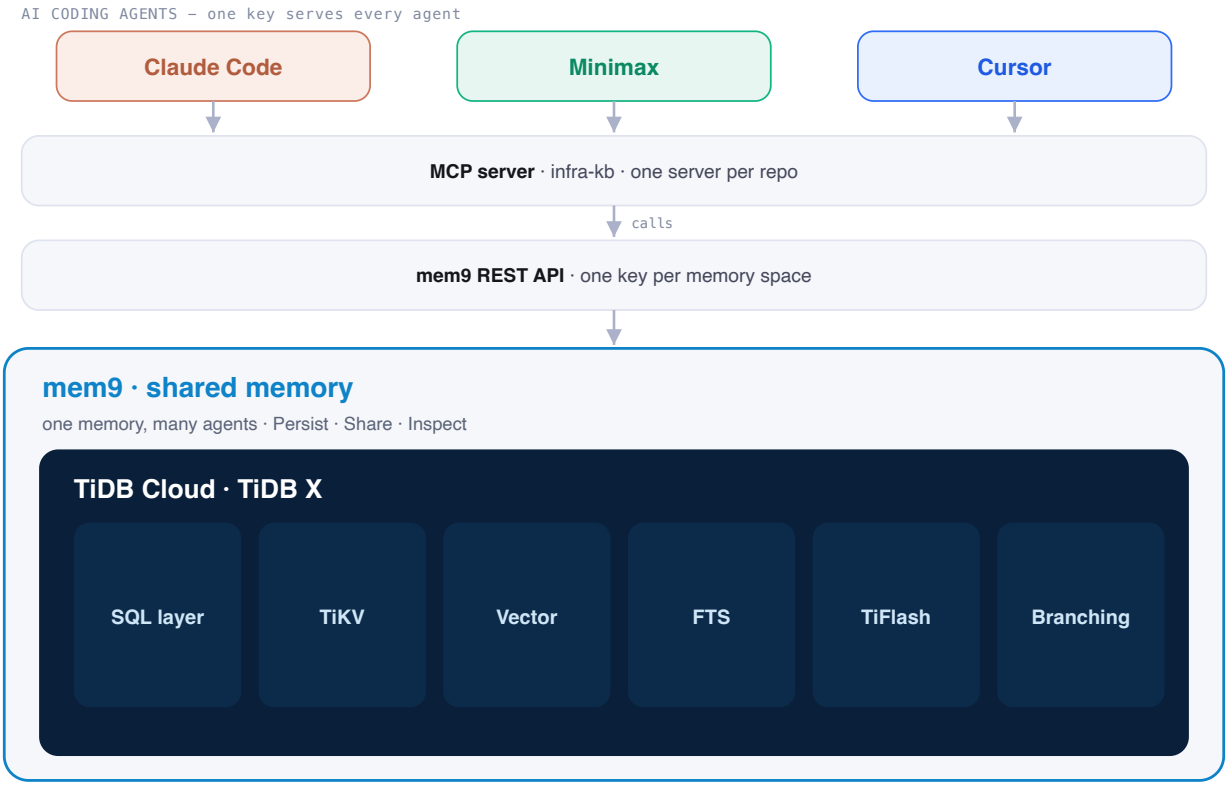
THE PROBLEM	HOW MEM9 SOLVES IT	SHOWN IN
Each team's knowledge must stay private - no team reading another's.	One **TiDB Cloud cluster per team** (a mem9 space). A team's key reaches only its own memory; isolation is enforced by the platform, not application code.	§ The model
One integration should serve many repositories, not one wiring per repo.	Within a team, repos are `appId` namespaces on one cluster and the agent **routes to the right one**. Across teams, separate keys mean separate clusters.	§ The model
A single task often spans repos - create a resource in one, the dependent resource in another.	One agent session writes to **each repo's namespace** and links the records by `account_ref` - one integration, two repos, no manual hand-off.	§ One task, two repos
Some teams require a self-hosted deployment.	**Supported.** Vector and graph run self-hosted; full-text search is TiDB Cloud only today.	§ Deployment

Demonstrated live with three different coding agents — Claude Code, Minimax, and Cursor — against a single TiDB Cloud cluster. The tools are interchangeable; the shared memory is the point.

ARCHITECTURE · MEM9 ON TIDB

mem9 is the memory. TiDB is the engine.

mem9 is the agent-facing product - **one memory, many agents**: a shared, persistent layer every coding agent reads and writes through a single key, with hybrid recall and a human-inspectable dashboard. An agent reaches it through two layers - its **MCP server**, which calls the **mem9 REST API**. And mem9 *is* TiDB Cloud at its core: the SQL layer, TiKV, vector and full-text search, TiFlash, and branching that store and serve every memory all run inside it - one engine, no separate vector database or sync layer.



An agent reaches mem9 through two layers - its MCP server, which calls the mem9 REST API. Inside mem9, a single TiDB Cloud engine holds the relational state, the dependency graph, vector embeddings, and full-text - the same store that computes the blast radius. One engine, no separate vector database, no ETL, no sync layer.

THE MODEL

Team = cluster. Repo = namespace.

mem9 maps onto how a team already works. A **team** is a mem9 **space** - one key, one isolated TiDB Cloud cluster. A **repo** is an `appId` namespace inside it, so repos share a cluster but keep separate recall scopes. Another team is another cluster, reachable only with its own key.

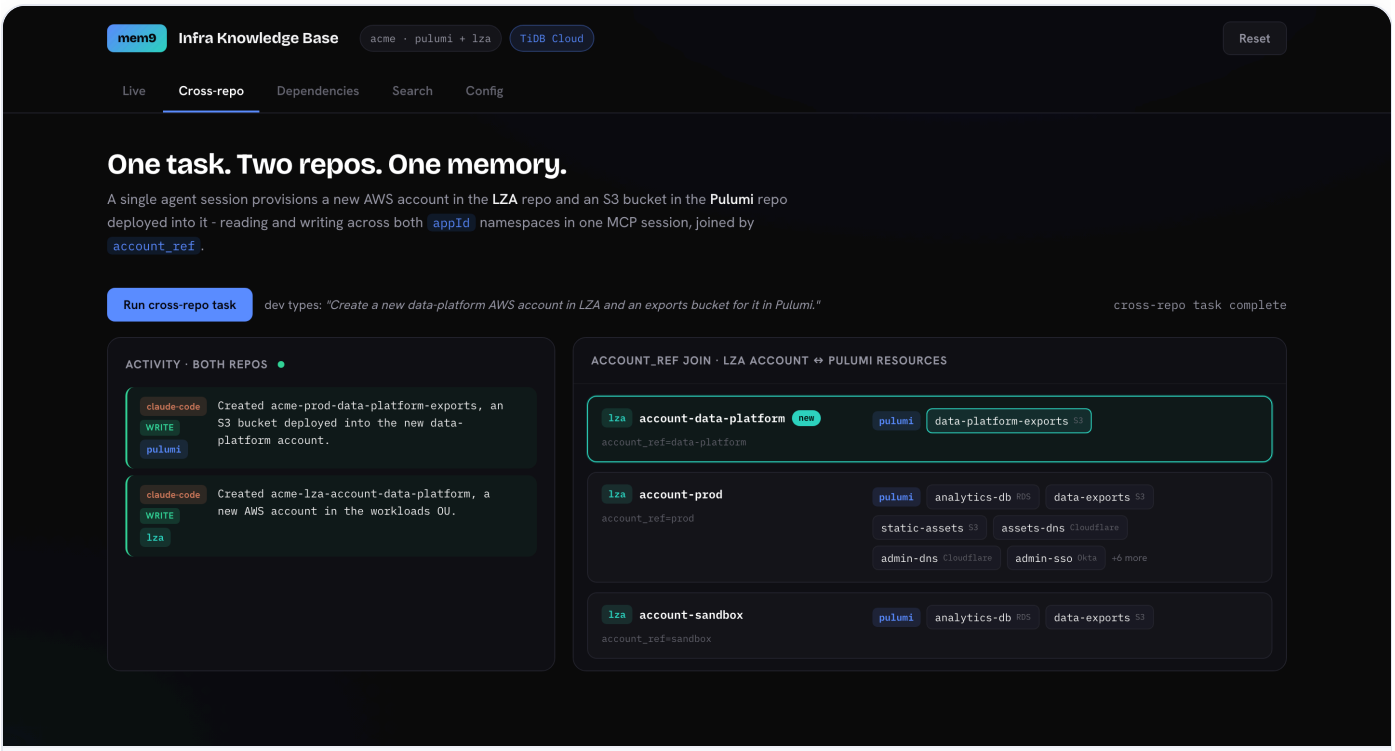
One key per team. Inside a team, repos are `appId` namespaces on the same cluster, joined by `account_ref`; across teams, separate keys mean separate clusters and zero shared recall.

Each team's space sits on its own managed TiDB Cloud cluster. Agents never handle raw database credentials – the mem9 API key is the only secret.

HEADLINE · ONE TASK, TWO REPOS

One task, two repos.

A single task often spans repos - create a new AWS account in the LZA repo *and* the S3 bucket for it in the Pulumi repo. One agent, given one plain-language request, writes the account to `acme_lza_kb` and the bucket to `acme_pulumi_kb` - two separate namespaces - and links them with `account_ref`. No second session, no manual hand-off.



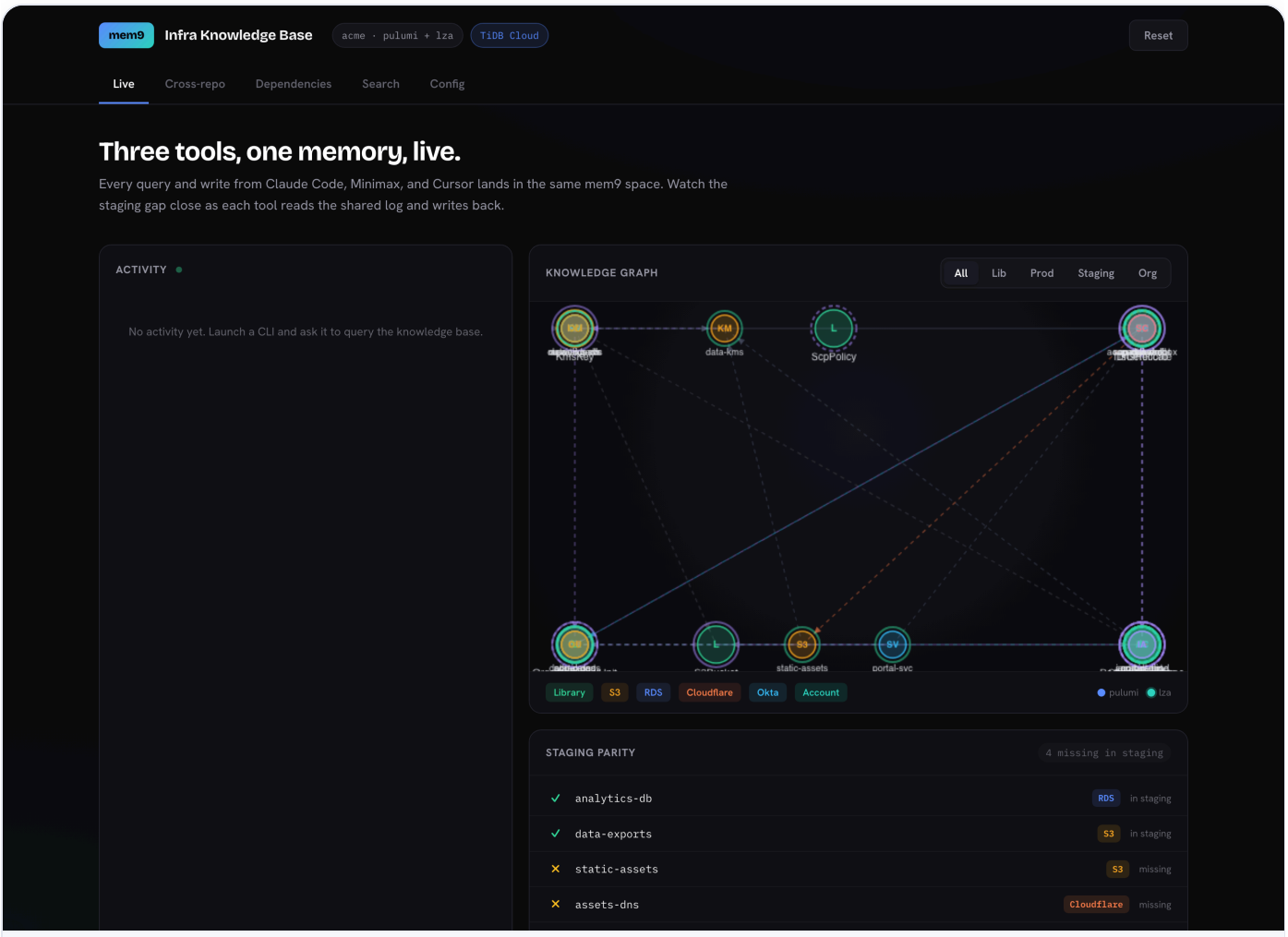
DASHBOARD · CROSS-REPO Left: one activity feed shows the same agent writing to **both** repos. Right: the `account_ref` join - the new **data-platform** account and its Pulumi exports bucket, created together and linked.

Why it matters. The agent doesn't need to be told which repo owns what. It routes each write to the right namespace and records the cross-repo link, so the next session - in either repo - already knows the account and its bucket belong together.

THE COLLABORATION

Three tools, one memory, warm hand-offs.

Production is complete; staging has drifted by four components. Three different agents bring it back to parity - the developer only states the goal. Every agent's rules file (`CLAUDE.md` / `AGENTS.md` / `.cursorrules`) makes the same first move non-negotiable: **query mem9 before writing anything**. So each step below runs the same shape - recall, compose, write back.



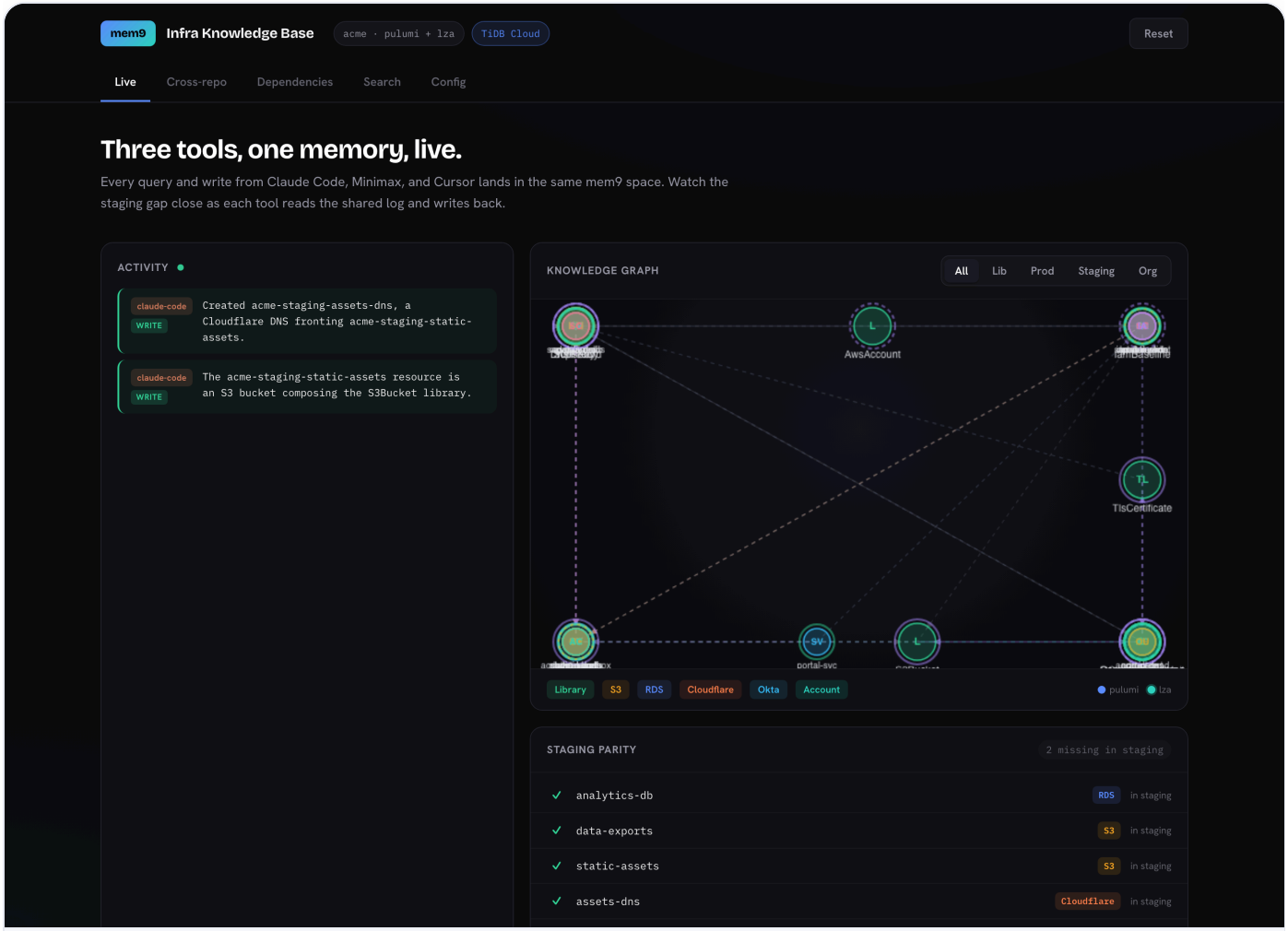
`STARTING POINT` Feed empty; the staging-parity strip shows **4 missing**.

STEP 1 Claude Code

```
dev types "Bring staging up to parity with production — start with the static-assets bucket and its CDN."
```


```
recall the gap → compose the S3 + DNS libraries → write both components back, with the dns-fronts-bucket edge.
```

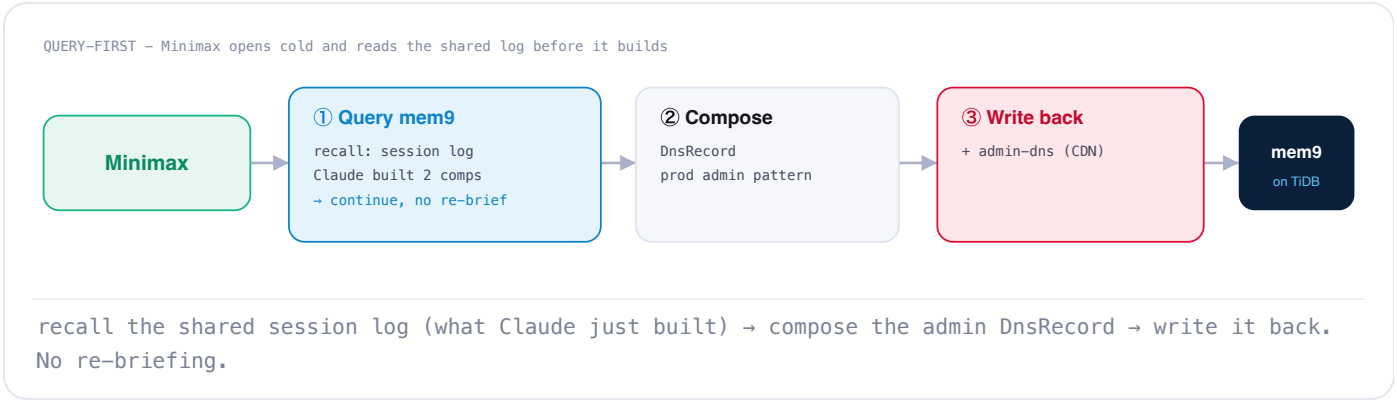
Result. Staging gains the **static-assets** bucket and its **CDN**. **Gap 4 → 2.**



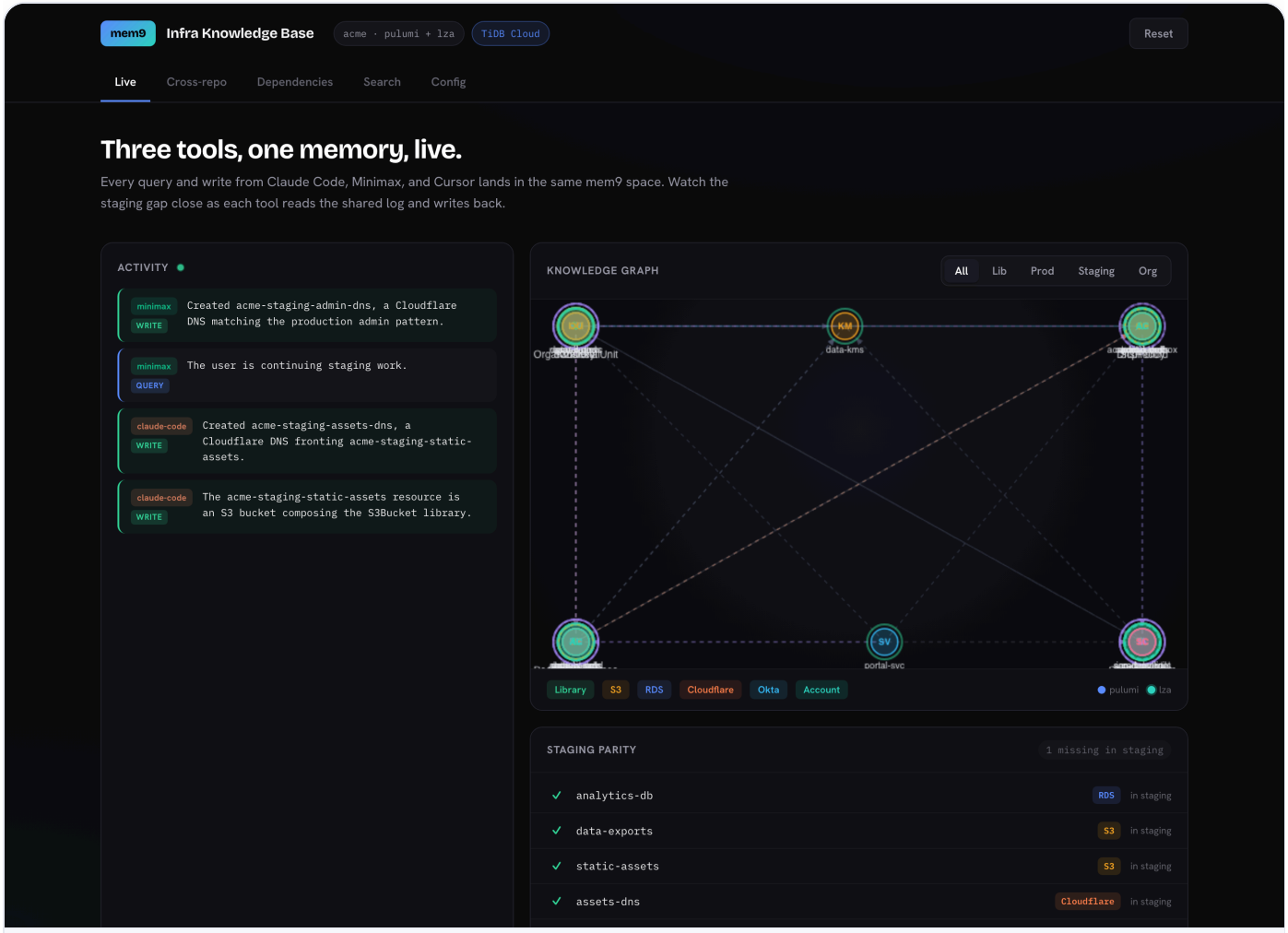
RESULT · CLAUDE CODE Both writes land in the shared feed; the parity strip drops to **2 missing**.

STEP 2 Minimax

```
dev types "Keep bringing staging to parity with prod — it still needs the admin-portal DNS."
```



Result. The admin-portal **DNS** lands - Minimax never had to be re-briefed. **Gap 2 → 1.**



RESULT · MINIMAX Its query ("continuing staging work") and write join the same timeline as Claude Code's. **1 missing**.

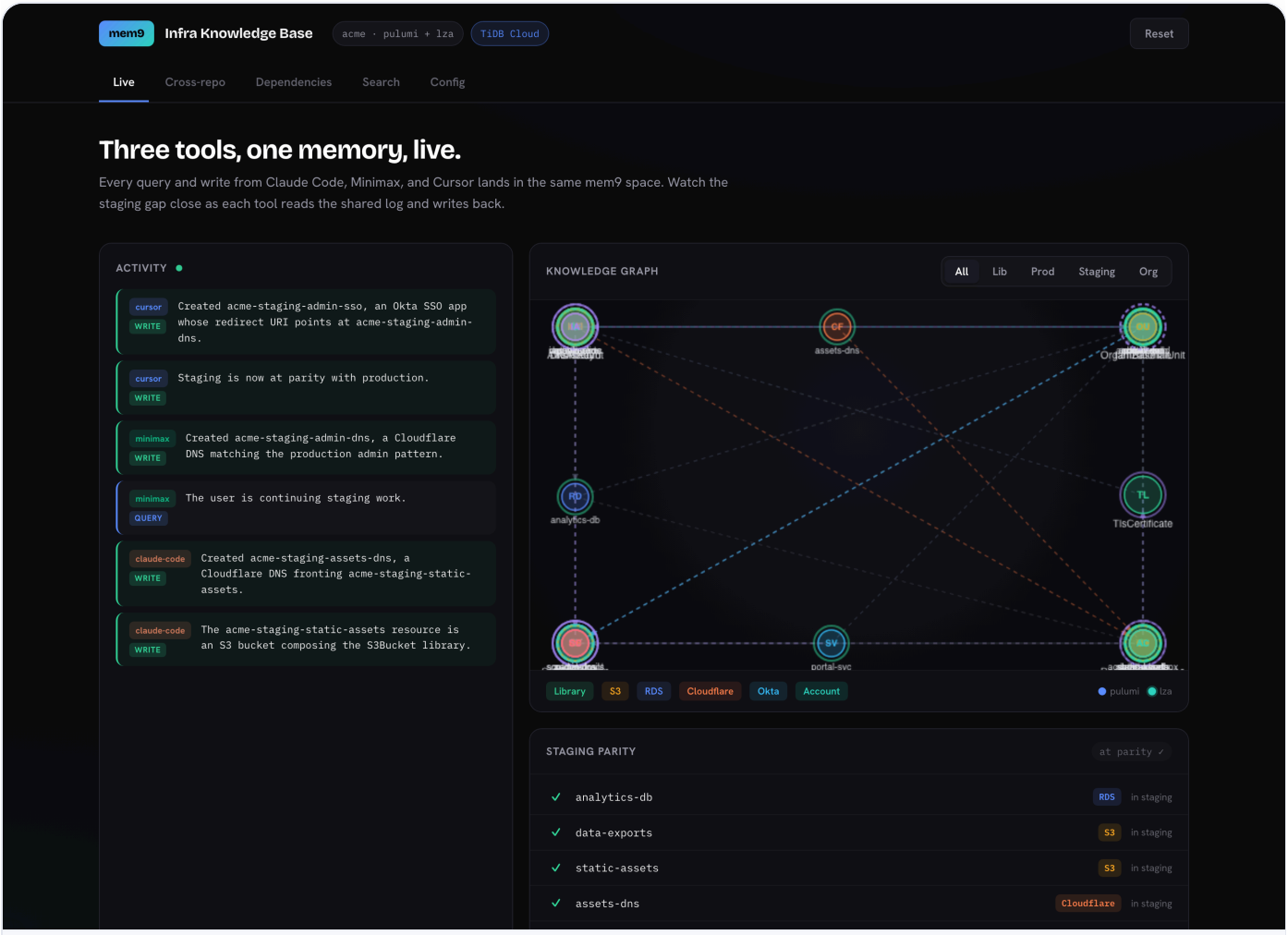
STEP 3 Cursor

```
dev types "Finish bringing staging to parity — add the admin SSO app."
```

QUERY-FIRST — Cursor traces the dependency chain before it builds


```
hybrid-search the prod SSO app → trace its depends_on / redirects_to chain → compose and write the matching staging SSO app.
```

Result. The SSO app's redirect URI points at the admin DNS; staging reaches **parity**. `Gap 1 → 0.`



RESULT · CURSOR One timeline now spans all three tools; the final write reads "Staging is now at parity with production." **0 missing**.

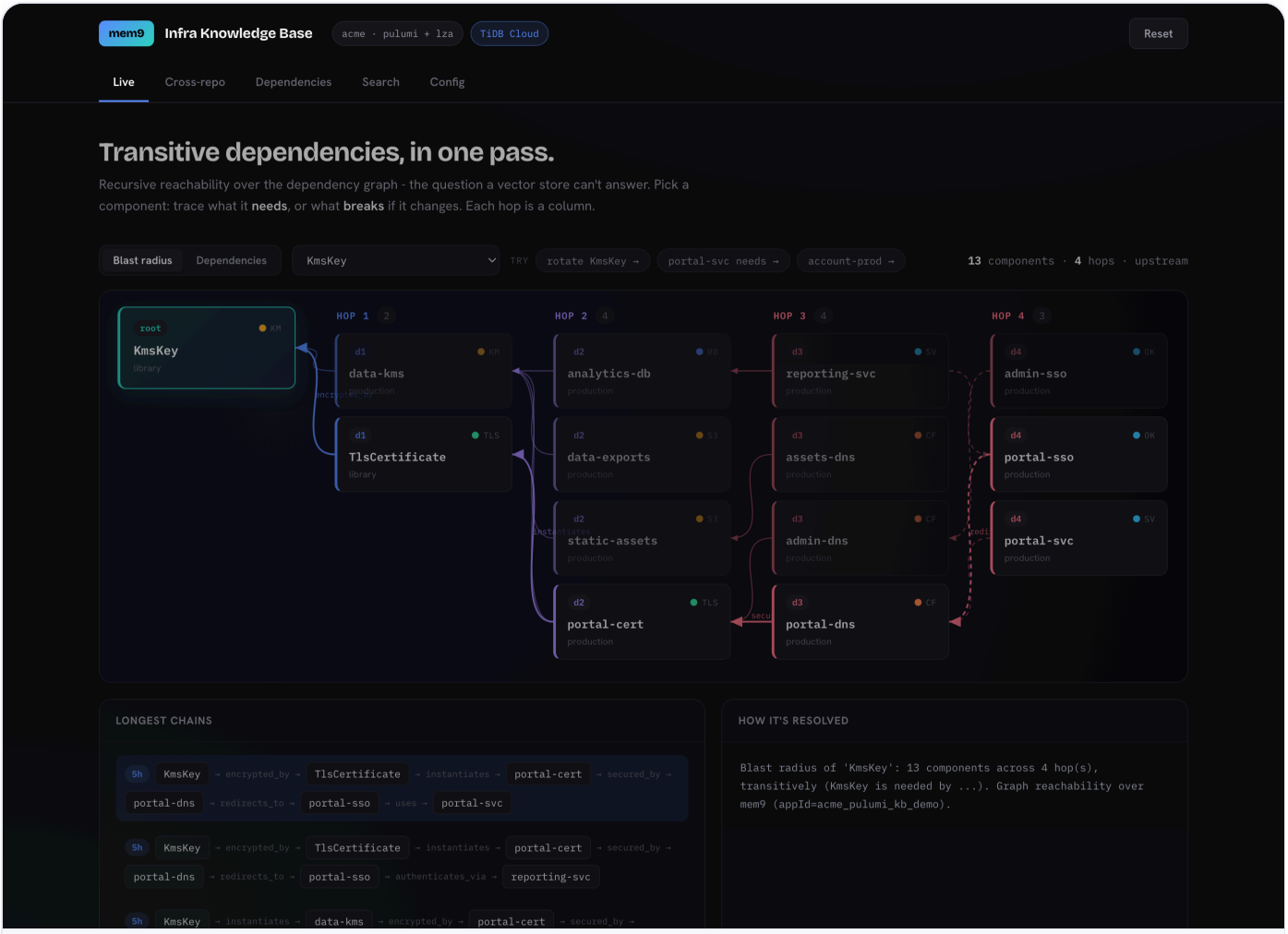
WHY TIDB

The question similarity can't answer.

A vector store finds things that *look* related. It cannot compute a transitive closure - "what, exactly, breaks if I change this?" That is graph reachability. mem9 records every `depends_on` and composition edge; one recursive traversal walks the chain to any depth.

`portal-svc` `authenticates_via` → `portal-sso` `redirects_to` → `portal-dns` `secured_by` → `portal-cert` `encrypted_by` → `data-kms` `instantiates` → **`KmsKey`**

Run that in reverse from `KmsKey` - "what breaks if we rotate it?" - and the answer is **13 components across four depths**. No flat search surfaces this; only walking the graph does.



DASHBOARD · DEPENDENCIES "What breaks if we rotate **KmsKey**?" Each hop is a column; 13 components fan across four depths and funnel onto the anchored foundation. The Trace panel reads each chain as a sentence.

One engine, not four

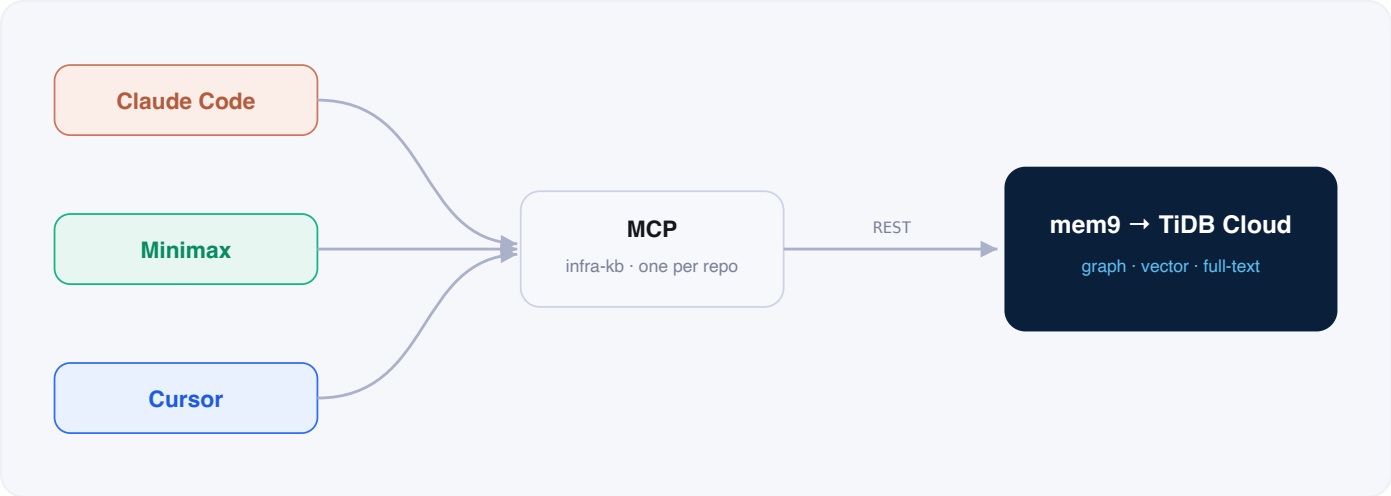
APPROACH	GRAPH REACHABILITY	SEMANTIC RECALL	ATOMIC AGENT STATE
`S3 / filesystem`	no traversal	no	no transactions
`Vector DB only`	similarity ≠ reachability	yes	no
`Four separate systems`	partial, with ETL	yes	no cross-system atomicity
`mem9 → TiDB Cloud`	recursive traversal	vector + full-text	one transaction

Graph, vector embeddings, and full-text live in the same store - the same recall that answers *"what encrypts production data"* also computes the blast radius above. No separate vector DB, no ETL, consistent across concurrent writes from all three tools.

> **S3 is the filing cabinet for artifacts. mem9 is the brain agents query.**

ARCHITECTURE & DEPLOYMENT

One store. One plug. One key.



MCP is the plug

Every tool loads the same Model Context Protocol config - one named server per repo. No custom glue per tool.

One store, every pattern

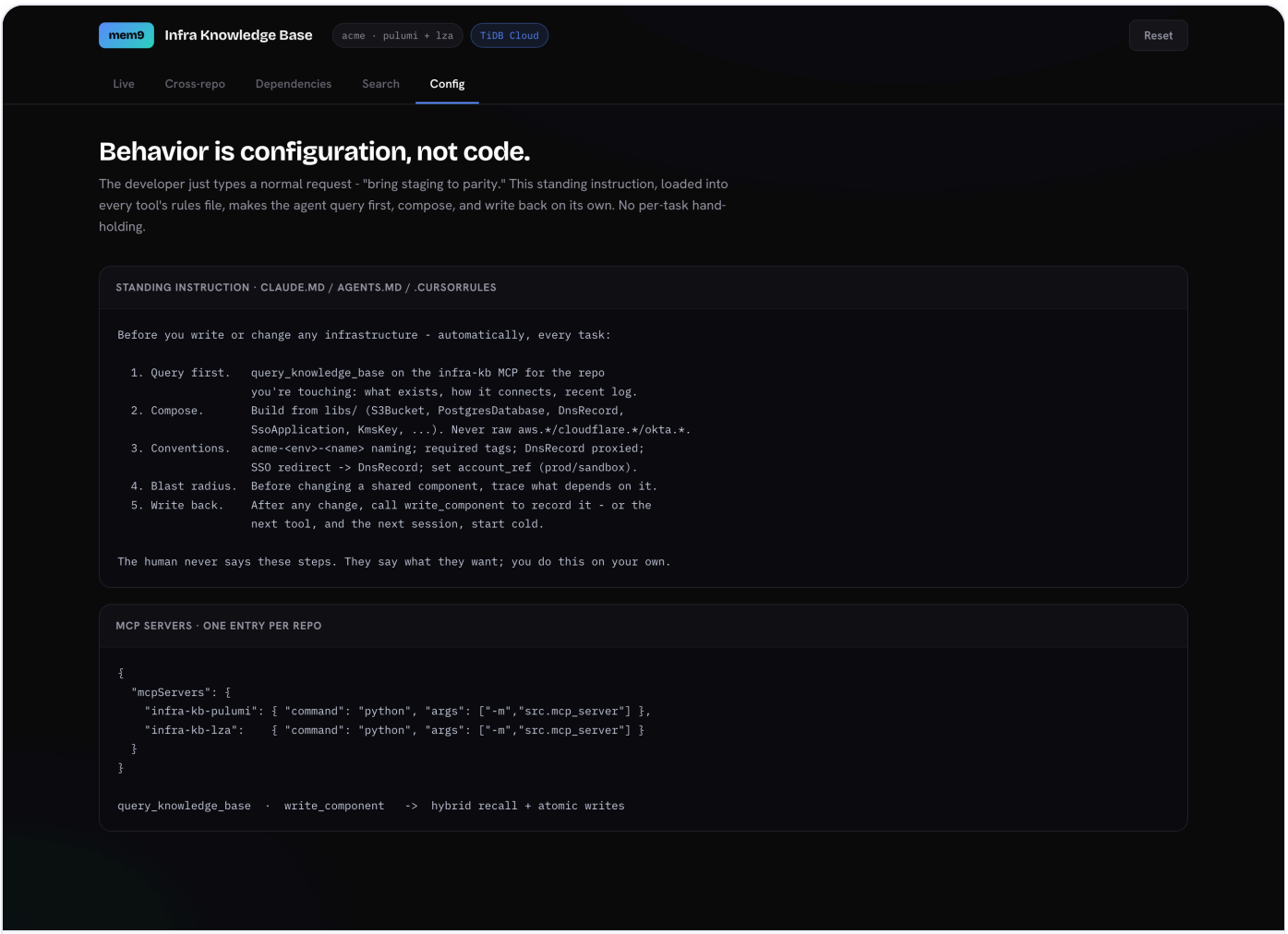
Relational state, the dependency graph, vector embeddings, and full-text all live in the same memory.

Just an API key

mem9 manages the TiDB Cloud cluster. Agents read and write through one key and never touch credentials.

The standing instruction

The developer types a normal request. The prescriptive part lives in each tool's rules file (`CLAUDE.md` / `AGENTS.md` / `.cursorrules`), so the agent queries first, composes, and writes back on every run with no per-task hand-holding.



DASHBOARD · CONFIG The five-step standing instruction every tool loads, and the one-entry-per-repo MCP wiring that makes it automatic.

Deployment

→ **TiDB Cloud (shown here):** graph + vector + full-text, fully managed; nothing to host.

→ **Self-hosted:** available - vector and graph run on-prem. Full-text search is TiDB Cloud only today, so this brief uses Cloud to show hybrid recall end to end.